

GEORGIOS KARAMOUSALIS

Software Engineer | Backend Development | Applied Artificial Intelligence

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PROFESSIONAL SUMMARY

Computer Science graduate and MSc Artificial Intelligence candidate with hands-on experience building full-stack applications, secure REST APIs and target-aware NLP systems. Strong foundation in Java, Python, JavaScript, SQL, relational databases, automated testing, Docker and GitHub Actions. Combines software engineering discipline with operational accuracy, customer service and research-led problem solving.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL | Backend: Spring Boot, FastAPI, Node.js, Express, REST APIs, JWT | Frontend: React, TypeScript, HTML, CSS | AI/Data: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NetworkX | Data/Platform: PostgreSQL, MySQL, MariaDB, Redis, RabbitMQ | Engineering: Git, GitHub Actions, Docker, JUnit, Pytest, Selenium, Swagger/OpenAPI

EDUCATION

MSc Artificial Intelligence | In progress

Focus: deep learning, natural language processing, model evaluation and applied machine learning research.

BSc Computer Science | Metropolitan College, Athens, Greece | 2022 - September 2025

Relevant areas: software engineering, databases, mobile and distributed systems, web development, data analysis and artificial intelligence.

SELECTED PROJECTS

Stance Detection with Deep Learning | MSc Thesis in Progress

- Built a reproducible target-text classification pipeline for FAVOR, AGAINST and NEUTRAL stance labels using VAST and EZ-STANCE datasets.
- Compared TF-IDF/Logistic Regression baselines with BERT and RoBERTa; best RoBERTa experiment achieved 80.3% accuracy and 80.3% Macro-F1 on EZ-STANCE.
- Implemented weighted loss, validation-based checkpoint selection, learning-rate scheduling, error analysis, confusion matrices and a local Streamlit demonstration application.

Restaurant Reservation App | Full-stack Web Application

- Developed React user flows for registration, login, restaurant browsing and reservation management.
- Implemented Node.js/Express REST endpoints, JWT authentication and MariaDB persistence for users, restaurants and bookings.
- Documented the API with a Postman collection and provided database setup instructions and sample data.

Social Network Analysis for Influencer Detection | BSc Thesis

- Analyzed social graph datasets with Python, NetworkX and Gephi to identify influential actors and communities.
- Applied PageRank, centrality measures and community detection, translating graph metrics into interpretable findings.

ADDITIONAL ENGINEERING PROJECTS

SupplyPilot - Operations Command Center | Java 21, Spring Boot, React, TypeScript, PostgreSQL

- Built an inventory, supplier, expense and purchase-order platform inspired by real retail workflows, with low-stock alerts and an interactive live demo.
- Engineered JWT-secured REST APIs with Flyway, Redis, RabbitMQ, Swagger, JUnit/MockMvc, Docker Compose, Prometheus and GitHub Actions CI/CD.

CareerFlow - Job Search Command Center | React, TypeScript, FastAPI, PostgreSQL, JWT, Docker

Built a responsive application tracker with resume-fit analysis and interview preparation, backed by a JWT-secured async FastAPI service, PostgreSQL, Pytest and Docker.

ADDITIONAL EXPERIENCE

Expenditure and Supply Office | Hellenic Army

- Supported expense records, supply documentation and administrative workflows requiring accuracy, organization and confidentiality.
- Coordinated routine office tasks and maintained dependable records in a structured, deadline-driven environment.

Retail Operations | Family Wine Shop

- Supported customer service, product handling, stock organization and daily retail operations.
- Built practical communication, commercial awareness and ownership through direct customer interaction.

LANGUAGES

Greek: Native | English: C2 / Proficient | German: Basic